

Mohamed Moslemani

AI/ML Engineer

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SUMMARY

AI/ML Engineer and Data Tech Lead with 2+ years of experience delivering production machine learning, LLM and agentic systems for consulting, banking and logistics clients. Currently leading a team of 5 at Strategy& (PwC Network), with a strong record of taking models from research to deployed services: forecasting, fraud detection, computer vision and LLM-based automation, backed by Databricks ETL pipelines, Kubernetes and full observability. MSc candidate in Computational Science (Machine Learning track) at the American University of Beirut on a full merit scholarship, with a BSc in Physics.

EXPERIENCE

AI/ML Engineer

Strategy& (PwC Network)

Jan 2026 – Present

Beirut, Lebanon (On-site)

- Data Tech Lead for a team of 5: set technical direction, review designs and code, and own delivery of the team's data and AI workstreams.
- Lead development of forecasting models for multiple business KPIs, with a primary focus on revenue modeling used for planning and decision support.
- Designed and delivered an end-to-end agentic AI system (LangGraph, FastAPI) and exposed it as an internal service, automating workflows across several departments.
- Lead the implementation of Databricks ETL pipelines; own and maintain 12 production pipelines with operational and observability data surfaced in Grafana and Prometheus.
- Ship containerized services on Kubernetes with Docker and Azure DevOps CI/CD, secured via Azure Entra ID; instrument LLM workloads with Langfuse for tracing and evaluation.
- **Tech:** Python (LangGraph, Langfuse, FastAPI, PyTorch, scikit-learn), Databricks, MS SQL Server, Azure DevOps, Azure Entra ID, Azure Databases, Docker, Kubernetes, Grafana, Prometheus, Jira (Agile).

AI Technical Consultant

QuanTech (Midis Group, IBM Partner)

Oct 2025 – Jan 2026

Beirut, Lebanon (On-site)

- Implemented an agentic system for financial use cases for Fransabank, built on LangChain and fast-agent and served through FastAPI.
- Built a credit card fraud detection system to flag fraudulent transactions using supervised machine learning.
- Developed enterprise ML solutions on IBM watsonx, deployed on Red Hat OpenShift and published through WSO2 API Manager.
- **Tech:** Python (LangChain, Langfuse, fast-agent, FastAPI, scikit-learn), IBM watsonx, OpenShift, WSO2 API Manager, Docker, Linux.

Data Scientist

800Storage

Feb 2024 – Jul 2025

Dubai, UAE (Remote)

- Built a real-time CCTV facial recognition system for automated personnel identification.
- Developed a volume estimation model with 93% accuracy, reducing operational costs by 25% and improving process efficiency by 30%.
- Fine-tuned LLMs for sales workflows, increasing conversions by 40%.
- Automated data cleaning, transformation, analysis and visualization pipelines, replacing manual reporting.
- **Tech:** Python, Java (Spring Boot), Docker, Kubernetes, AWS, Redis, Linux.

Data Intern

UNICEF (United Nations)

Sep 2023 – Feb 2024

Lebanon (On-site)

- Validated geospatial and operational data across 20+ project sites.
- Improved data accuracy by 15% through automated Python and pandas workflows.
- Increased student engagement by 20% through data-driven platform enhancements.

EDUCATION

MSc in Computational Science, Machine Learning Track

American University of Beirut

2026

Beirut, Lebanon

- Graduate Fellowship and Assistantship Program: full merit scholarship.
- Teaching Assistant: CMPS 262 Data Science in R and Python; CMPS 208 Business for Computing.
- Thesis (in progress): distribution shift and generalization of models trained on one data distribution to different

test distributions.

- Coursework: Statistical Learning, Large Language Models, Reinforcement Learning, Non-linear Optimization, Algorithmic Graph Theory, Partial Differential Equations, AI in Industry.

Graduate Diploma in Artificial Intelligence and Data Science

2024

American University of Beirut

Beirut, Lebanon

- Coursework: Data Science, Machine Learning, Deep Learning, Business Analytics, Arabic Natural Language Processing.

BSc in Physics

2023

Beirut Arab University

Beirut, Lebanon

- Top Student of the Department (2022–2023); Faculty of Science Representative on the University Council (2022–2023).
- Coursework: Linear Algebra, Multivariate Calculus, Complex Analysis, Quantum Mechanics I & II, Solid State Physics, Electronics, Java for Computing.

SKILLS

Programming: Python, R, Julia, Java, JavaScript, SQL

AI / ML: PyTorch, TensorFlow/Keras, scikit-learn, Transformers, LLMs, RAG, LLM fine-tuning, CNNs, Vision Transformers, Autoencoders, 3D Computer Vision, OCR, Time-series Forecasting, Reinforcement Learning

LLM / Agents: LangGraph, LangChain, Langfuse, fast-agent, IBM watsonx

Data & Databases: Databricks, Spark, Airflow, pandas, PostgreSQL, MS SQL Server, MongoDB, Redis, FAISS, BM25

MLOps & Cloud: Docker, Kubernetes, OpenShift, Azure (DevOps, Entra ID), AWS, MLflow, GitHub Actions, CI/CD, WSO2 API Manager

Backend & Tools: FastAPI, Flask, Spring Boot, React, Linux, Git, Grafana, Prometheus, Alertmanager, Jupyter, Plotly, Matplotlib, Seaborn

PERSONAL PROJECTS AND RESEARCH

DocFlows: Enterprise Document Workflow Automation Platform

docflows.com | github.com/Mohamed-Moslemani/MainProjectRepo

- Architected a distributed document automation platform using microservices, asynchronous task queues and workflow orchestration for end-to-end document lifecycle management.
- Built OCR and computer vision pipelines for document understanding: image preprocessing, information extraction, schema validation and error recovery across heterogeneous document formats.
- Set up production infrastructure with Docker, CI/CD, centralized logging, distributed tracing and metrics-based monitoring.

Cancer Type Classification on X-Ray Images

github.com/Mohamed-Moslemani/cancer-multiclass-classification

- Built a multiclass cancer classifier with few-shot transfer learning on a limited medical X-ray dataset.
- Combined InceptionV3 and ResNet50 backbones in a hybrid architecture with a custom dense head; achieved state-of-the-art results on the target datasets with a reproducible TensorFlow/Keras pipeline.

3D Voxel Reconstruction with Transformer Architectures

- Designed a transformer-based model for 3D voxel completion from partial or degraded inputs, operating directly on raw STL mesh data.
- Achieved high-fidelity reconstructions with stronger generalization than baseline 3D autoencoder models.

Numerical Simulation of Fourier and Non-Fourier Heat Conduction

- Implemented and benchmarked numerical simulations comparing classical Fourier conduction against Green–Naghdi non-Fourier models under transient thermal conditions, addressing the infinite propagation speed paradox.